

Comprehensive Sensitivity Analysis Report

Week 7 evaluates how the JPM chooser model and the original selected machine-learning routes respond to volatility, interest rates and VIX. The required shocks increase the structural BSM price, while the ML routes respond differently. The analysis also records the complete-feature update and fallback controls used by the prototype.

Reference Contract and Observation

Input	Value
Observation	2024-11-29, last complete Week 5 target row
JPM spot and strike	\$241.1194 and \$150
Choice and maturity	T1 = 0.5 year; T2 = 1 year
Volatility and rate	45.7249% annual volatility; 4.18% Treasury proxy
Dividend yield	Paper assumption q = 2.33%

Structural BSM Shocks

The volatility shock multiplies annualized volatility by 1.5. The rate shock adds 0.02 to the decimal rate, taking 4.18% to 6.18%; it is an absolute two percentage-point rise. Spot, strike, dividend yield and contract dates remain fixed. The combined scenario applies both changes.

Scenario	BSM price USD	Change USD	Change percent
Base	99.2617	+0.0000	+0.00%
50% Volatility Spike	115.6683	+16.4066	+16.53%
2% Rate Hike	101.3317	+2.0700	+2.09%
Combined Shock	116.9177	+17.6561	+17.79%

The volatility shock increases the chooser price by \$16.4066, compared with \$2.0700 for the rate shock. These are finite scenario changes at the stated contract, not universal sensitivities or observed trading profits. The combined change is not assumed to equal the sum of separate shocks.

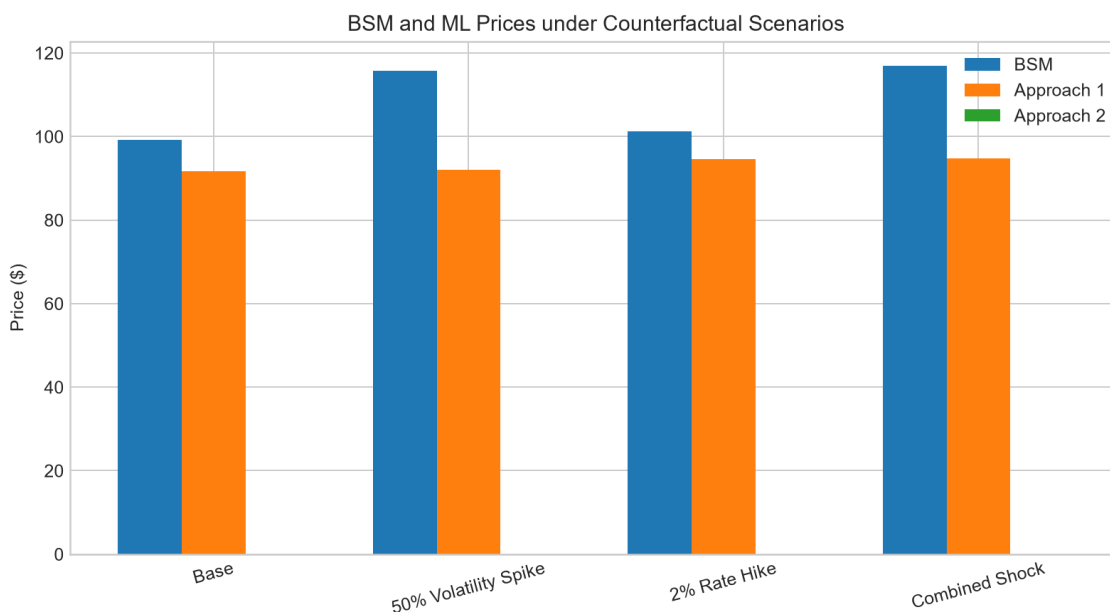
Machine Learning Counterfactuals

The original Week 6 Random Forest predicts forward volatility for Approach 1, which is passed through BSM. The original Linear Regression predicts the proxy price directly for Approach 2. No dividend-growth feature or model retraining is introduced by this analysis.

For the ML volatility scenario, all rolling-volatility features and VIX level are multiplied by 1.5, with VIX return set to at least 0.50. The rate scenario adds 0.02 to the decimal Treasury feature and 2.0 to its percentage-point momentum. Other features stay fixed. These coordinated feature shocks differ from a one-input BSM sensitivity curve.

Scenario	BSM USD	RF plus BSM USD	Direct price USD
Base	99.2617	91.7072	0.0000
50% Volatility Spike	115.6683	92.0449	0.0000
2% Rate Hike	101.3317	94.5557	0.0000
Combined Shock	116.9177	94.8010	0.0000

Figure 1 Original model counterfactual prices



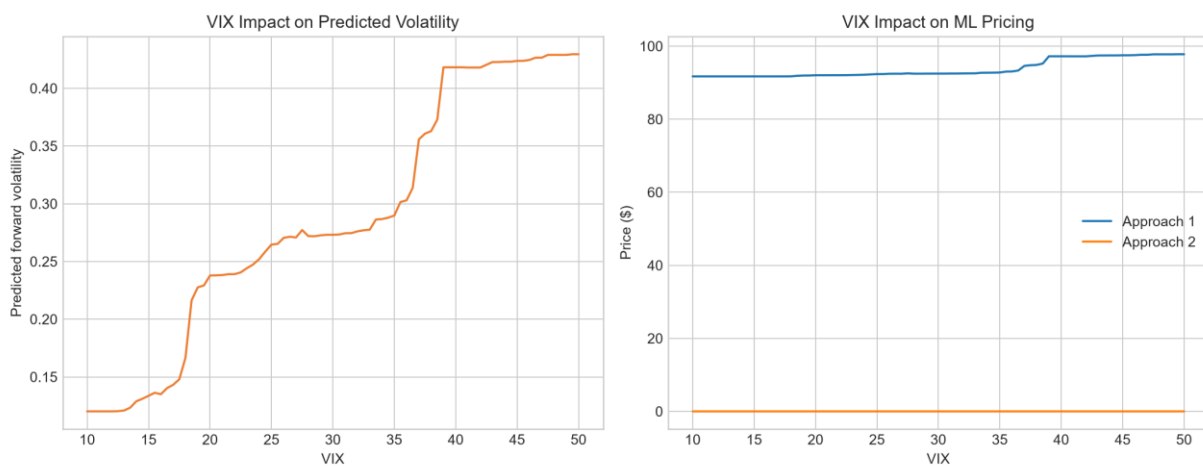
The RF route rises by only \$0.3378 under the volatility shock because its predicted forward volatility changes differently from the imposed historical-volatility input. Direct-model outputs are clipped at zero by the implementation. A flat zero output is a model limitation, not evidence that the option is worthless or insensitive.

VIX Sensitivity and Model Interpretation

Explained model	VIX share of mean absolute SHAP
Approach 1 Random Forest	40.80%
Approach 2 Linear Regression	5.06%

The SHAP shares explain the exact selected models on the analyzed sample. They measure fitted reliance on VIX-related features, not a causal effect or a news-sentiment contribution. The Random Forest route assigns a larger share to these features than the direct linear route.

Figure 2 VIX level counterfactual on the reference observation



The saved VIX grid spans 10.0 to 50.0. Only VIX level varies in this grid; other features retain their reference values. Approach 1 prices range from \$91.7071 to \$97.7777. Tree thresholds can produce flat segments and jumps, so this curve should not be read as a smooth Greek.

Counterfactual rows can lie outside observed feature combinations. The fixed \$150 strike is also far below the reference spot. These constraints matter when interpreting both weak price responses and extrapolation. VIX remains a volatility indicator; the unavailable news-sentiment series is not synthesized or substituted.

Complete Feature Updates and Failure Controls

The Week 7 and Week 8 tools preserve the original 17 volatility features and 19 direct-price features. JPM OHLCV, VIX and DGS10 histories rebuild returns, rolling volatility, moving-average gaps, momentum, correlations and trading-activity features at a common completed observation date. Live calculations are checked against the archived Week 5 features. The models and research sample remain unchanged.

Control	Implementation
Request bounds	12-second timeout and at most two attempts per request
Source independence	Parallel requests and a separate atomic JSON cache per source
Missing source	Use its last available history; label cached or unavailable status
Incomplete feature row	Retain the prior complete snapshot rather than combine current spot with old features
Rate meaning	DGS10 stays the trained ML feature; DGS1 supplies the one-year BSM pricing rate
Automatic refresh	On opening and every 15 minutes during an active app session; manual refresh remains available
Credentials	Read FRED key from environment or local .env; omit keys and authenticated URLs from logs

On 10 September 2026, both updater paths returned all four sources successfully and produced a complete feature row dated 8 September 2026, the latest common date. Tests also covered source timeout, per-source fallback, corrupt cache and historical feature parity. Provider delays can still occur; a successful update is not a guarantee of intraday freshness.

Conclusion and Evidence

The required volatility and rate stresses, original-model counterfactuals, VIX analysis and complete update controls are delivered. News sentiment and actual CME chooser transactions remain disclosed limitations. The additional daily GitHub refresh workflow is prepared locally and becomes active after the revised project is published.

Evidence: Week7_Advanced_Analysis.ipynb; sensitivity_analysis.py;
model_results/structural_stress_scenarios.csv; ml_counterfactual_scenarios.csv;
shap_feature_group_impact.csv; vix_counterfactual_curve.csv; .github/scripts/test_feature_refresh.py. All prices are model estimates in USD and are not trading recommendations.